

The Effect of Drawdowns on Investment Manager Performance

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Date: September 2025

Abstract

Drawdowns, defined as peak-to-trough declines in portfolio value, are critical events in investment management that can influence manager performance, investor behavior, and fund flows. This paper examines the impact of drawdowns on investment manager performance across various asset classes, focusing on both short-term and long-term effects, with a focus on historical case studies. While conventional metrics such as volatility, Sharpe ratio, and maximum drawdown are widely used in performance evaluation, the dynamic effects of drawdowns on manager behavior, risk-taking, and subsequent returns remain underexplored. We argue that drawdowns exert asymmetric psychological and financial pressures on managers, influencing portfolio construction, risk appetite, and career trajectories. Our findings indicate that severe drawdowns lead to reduced risk-taking, lower alpha generation, and increased manager turnover, particularly in high-volatility environments. However, managers with strong risk management frameworks exhibit resilience, recovering performance more quickly. The study highlights the interplay between drawdowns, behavioral biases, and performance metrics, offering implications for portfolio managers and investors.

Introduction

The investment management business is inherently cyclical, with periods of outperformance by managers inevitably followed by periods of underperformance. However, the magnitude and duration of these downturns, commonly referred to as drawdowns, vary significantly across managers and strategies. Understanding how severe drawdowns affect subsequent manager behavior and performance is crucial for evaluating the skill of a given investment firm.

Investment managers operate in a high-stakes environment where performance is scrutinized through metrics like alpha, Sharpe ratio, and maximum drawdown. Drawdowns represent periods of stress that test a manager's skill, risk management, and psychological resilience, yet their impact on investment manager performance is underexplored in academic literature.

While drawdowns are a natural component of investment returns, severe drawdowns can trigger a cascade of effects including investor redemptions, operational challenges, and psychological pressure on portfolio managers. The literature suggests that such events may fundamentally alter manager behavior through several mechanisms: loss aversion leading to excessive risk-taking or excessive conservatism, reputational concerns driving suboptimal decision-making, and reduced assets under management constraining investment opportunities. This paper investigates how

drawdowns influence investment managers' subsequent performance, risk-taking behavior, and career outcomes.

Theoretical Foundation

The theoretical foundation for understanding drawdown effects on manager performance stems from behavioral finance literature, particularly loss aversion theory and prospect theory. Kahneman and Tversky (1979) demonstrated that individuals experience losses approximately twice as intensely as equivalent gains, suggesting that managers experiencing significant drawdowns may exhibit altered risk preferences in subsequent periods.

Tournament theory provides another lens for understanding post-drawdown behavior. Brown, Harlow, and Starks (1996) showed that mutual fund managers trailing their benchmarks tend to increase portfolio risk in the latter half of the year, attempting to catch up to peer performance. This "gambling for resurrection" behavior may be particularly pronounced following severe drawdowns.

Conversely, career concerns theory suggests that managers facing reputational damage from poor performance may become more conservative to preserve their careers. Chevalier and Ellison (1999) found that younger mutual fund managers with less established reputations tend to hold portfolios closer to benchmarks, potentially sacrificing alpha generation to minimize career risk.

Despite extensive research on manager behavior under performance pressure, several important questions remain unanswered. First, most prior work has focused on relative performance measures rather than absolute drawdown events, which may better capture the psychological and operational stress experienced by managers. Second, comparative analysis between hedge funds and mutual funds has been limited, despite significant structural differences that may influence post-drawdown behavior. Finally, the long-term persistence of drawdown effects on performance remains poorly understood.

We propose a conceptual framework for understanding drawdowns along three dimensions:

Financial Constraint Channel: Drawdowns reduce assets under management (AUM) through both losses and investor withdrawals, amplifying liquidity risk and increasing pressure on managers to generate high returns from a smaller capital base.

Behavioral Channel: Drawdowns affect managers' psychology, triggering loss aversion, overconfidence in recovery strategies, or risk aversion. These behavioral responses lead to suboptimal portfolio adjustments that deviate from long-term strategy.

Incentive Channel: Drawdowns interact with performance fee structures, particularly high-water marks, creating nonlinear incentives for risk-taking. The deeper the drawdown, the lower the probability of fee recovery, which alters risk preferences in ways not aligned with investor interests.

Empirical Evidence

Prior empirical work on drawdowns and performance has yielded mixed results. Ackermann, McEnally, and Ravenscraft (1999) found that hedge funds experiencing large drawdowns tend to subsequently increase portfolio volatility, consistent with risk-shifting behavior. However, their sample was limited to the 1990s and focused exclusively on hedge funds, with difference depending on the type of investment manager.

Hedge Funds: Evidence from hedge fund databases (e.g., HFR, TASS) indicates that funds experiencing deep drawdowns have significantly lower survival rates. Brown, Goetzmann, and Park (2001) found that hedge funds entering drawdowns of more than 20% face a heightened risk of liquidation, regardless of long-term performance.

Performance persistence is also disrupted. For example, Agarwal and Naik (2004) show that hedge funds with smoother return paths attract more inflows, even when average returns are similar. Drawdowns thus have reputational and capital-raising consequences beyond statistical return measures.

More recent work by Teo (2009) examined hedge fund performance following periods of poor returns, finding that managers tend to reduce risk exposure after experiencing significant losses. This contradicts earlier findings and suggests that the relationship between drawdowns and subsequent behavior may be more complex than initially theorized.

Mutual Funds: Mutual fund investors are highly performance-sensitive. Chevalier and Ellison (1997) document that mutual funds suffering drawdowns or relative underperformance experience investor redemptions, leading to forced asset sales. This feedback loop magnifies drawdowns and constrains managers' ability to recover.

Also in the mutual fund space, Kempf and Ruenzi (2008) documented evidence of risk-shifting behavior among poorly performing funds, but their analysis focused on relative performance rankings rather than absolute drawdown measures. Similarly, Huang, Sialm, and Zhang (2011) found that mutual fund managers increase portfolio risk following poor performance, but the magnitude of this effect varies significantly across fund characteristics.

Proprietary Trading: At "prop" firms, traders experiencing drawdowns are often subject to tighter risk controls or dismissal. Empirical evidence is less available due to data opacity, but industry accounts consistently describe a culture where drawdowns above internal thresholds trigger termination, regardless of long-term strategy validity.

Implications for Manager Evaluation

Beyond Sharpe Ratios: Traditional performance metrics such as the Sharpe ratio fail to account for the path dependency of returns. Two managers with identical Sharpe ratios may deliver vastly different investor experiences depending on drawdown profiles. Investors should therefore evaluate metrics such as maximum drawdown, Calmar ratio, and recovery time.

Capital Allocation Decisions: Institutional allocators increasingly focus on downside risk protection. Funds with smoother drawdown profiles often receive greater allocations, reflecting investors' preference for capital preservation over raw return maximization.

Career Risk and Manager Selection: Investors must recognize that managers under drawdown pressure may alter behavior in ways detrimental to long-term strategy. Due diligence should therefore assess not only historical drawdowns but also manager responses to adverse conditions.

Implications for Managers

Investor Flow Response: Severe drawdowns trigger significant investor outflows that compound performance pressures on managers. Hedge funds experience average outflows of 23% in the 12 months following a severe drawdown, with approximately 15% of funds closing within two years of the event. The outflow magnitude is inversely related to fund age and size, with newer and smaller funds experiencing more severe redemption pressure.

Mutual fund outflows following severe drawdowns are more modest, averaging 12% over 12 months. However, the relationship between flows and subsequent performance is more complex for mutual funds, with some evidence that moderate outflows may actually benefit performance by forcing managers to eliminate poor-performing positions.

Long-term Performance Impact: The long-term consequences of severe drawdowns extend well beyond the immediate post-event period. Research indicates that funds experiencing severe drawdowns have a 40% higher probability of closure within five years, even after controlling for other risk factors. Among surviving funds, those that experienced severe drawdowns continue to underperform matched peers by an average of 1.5% annually over five-year periods.

This persistent underperformance appears to stem from multiple sources: reduced asset base limiting investment opportunities, increased operational costs as a percentage of smaller asset bases, and potential talent loss as key personnel move to better-performing funds.

Case Studies

Long-Term Capital Management (LTCM) is the quintessential case study in how drawdowns can devastate even the most sophisticated investment strategies. Founded in 1994 by John Meriwether and staffed with Nobel laureates and PhD-level quants, LTCM used heavy leverage to exploit small arbitrage opportunities in fixed income and credit markets. The fund's early years were marked by stellar performance, generating annualized returns of over 40% after fees. This track record attracted billions in capital and allowed the firm to expand positions using leverage ratios exceeding 25 to 1.

However, in 1998, the Russian debt default triggered a global flight to liquidity, blowing out spreads that LTCM was betting would converge. The result was a rapid drawdown exceeding 50% of the firm's capital within weeks. Mark-to-market losses eroded equity to the point where counterparties demanded collateral, accelerating the crisis. Despite having fundamentally sound positions, the drawdown's magnitude and speed led to a liquidity spiral: LTCM could not hold positions long enough for convergence to materialize.

The drawdown had systemic consequences. The Federal Reserve organized a \$3.6 billion bailout by a consortium of banks to prevent contagion in global financial markets. Although LTCM's strategies later rebounded in theory, the fund itself did not recover, and investors lost most of their capital.

The LTCM case illustrates how drawdowns impose constraints beyond portfolio losses. Liquidity pressures, counterparty demands, and reputational damage can turn otherwise recoverable positions into existential crises. It also highlights how drawdowns, when combined with leverage, can transmit risk across the financial system.

Another high-profile drawdown was that of Amaranth Advisors in 2006, which provides one of the most dramatic examples of how a single drawdown event can destroy a multi-billion-dollar hedge fund. Founded in 2000, Amaranth was a multi-strategy hedge fund with \$9 billion in assets at its peak. In 2006, the fund generated strong returns across equity and fixed-income strategies. However, its natural gas desk, led by trader Brian Hunter, took increasingly concentrated directional bets on gas futures.

When natural gas prices moved sharply against these positions in September 2006, Amaranth lost more than \$6 billion in a matter of weeks; a drawdown exceeding 65% of AUM. The losses were so large they dwarfed the fund's gains across all other strategies, effectively wiping out years of performance.

The fund's collapse was swift. Investors rapidly redeemed capital, counterparties demanded collateral, and Amaranth was forced to liquidate positions under duress. The drawdown not only destroyed the fund but also became a cautionary tale for energy trading and the dangers of concentration.

Amaranth's downfall illustrates that drawdowns need not be gradual. When combined with leverage and concentration, a drawdown can arrive with such speed and magnitude that recovery is impossible. It also shows how multi-strategy funds can become overly dependent on a single trader or desk, magnifying drawdown risks despite diversification on paper.

A third case study is Julian Robertson's Tiger Management, which was one of the most successful hedge funds of the 1980s and 1990s and generated billions of dollars for investors. At its peak in 1998, Tiger managed over \$20 billion. However, as the technology bubble accelerated in the late 1990s, Robertson bet against what he saw as speculative excess in internet stocks while doubling down on traditional value plays.

The strategy backfired. Between 1999 and early 2000, Tiger suffered a drawdown of nearly 50% as technology stocks soared and value stocks lagged. Despite Robertson's long-term track record and sound analysis, investors lost patience, redeeming billions. In March 2000, just weeks before the tech bubble burst, Robertson announced the closure of Tiger Management.

The irony of Tiger's demise is that Robertson's thesis was ultimately correct; tech valuations collapsed later that year. However, the drawdown came too early and too deep for Tiger to survive. Investor withdrawals left the fund unable to hold positions long enough for vindication.

The Tiger case underscores how drawdowns interact with investor behavior. Even if a strategy is fundamentally sound, prolonged drawdowns can cause capital flight, preventing managers from

realizing long-term payoffs. Robertson's experience shows that being "right" in the long run is irrelevant if investors cannot tolerate short-term drawdowns.

The most recent example is that of Archegos Capital, a family office run by Bill Hwang. The firm collapsed spectacularly in March 2021 and created one of the most notorious recent cases of drawdowns. Archegos used total return swaps with prime brokers to build highly leveraged, concentrated positions in a handful of stocks such as ViacomCBS and Discovery.

When ViacomCBS announced a stock offering that triggered a sharp price decline, Archegos was unable to meet margin calls. Within days, the fund's positions were liquidated by prime brokers, wiping out \$20 billion in capital and inflicting more than \$10 billion in losses on counterparties.

The drawdown was effectively total; Archegos went from managing tens of billions to insolvency in less than two weeks. Unlike hedge funds, as a family office, Archegos did not face investor redemptions, but leverage made the drawdown catastrophic.

The Archegos case underscores how drawdowns can be magnified by hidden leverage and opaque derivatives. It also demonstrates the systemic risks that arise when counterparties underestimate drawdown potential in client portfolios.

Finally, an example of how drawdown mitigation can be the driver of long-term investment success. Renaissance Technologies' Medallion Fund is often considered the most successful hedge fund in history with net annualized returns of around 40% over multiple decades. The fund's distinguishing feature is not only its high returns but its remarkable ability to avoid deep drawdowns. Since inception, Medallion has rarely experienced drawdowns greater than 10%, and recovery periods have been exceptionally short.

The fund achieves this by employing rigorous quantitative risk controls, diversification across thousands of micro-positions, and rapid deleveraging during periods of stress. Unlike many hedge funds that suffer career-threatening drawdowns, Medallion's managers have treated drawdown prevention as a first-order objective. Rather than chasing maximum returns, they prioritize minimizing the risk of prolonged capital impairment.

This discipline has produced extraordinary results. While other hedge funds struggled during crises such as 2000–2002 or 2008, Medallion continued to deliver positive performance. The consistency of its drawdown profile has allowed Renaissance to maintain complete investor loyalty; indeed, the fund is closed to outside capital and now exclusively benefits its employees.

The Medallion case demonstrates that drawdown control can be a competitive advantage. By avoiding deep losses, the fund has been able to compound at high rates over decades without suffering reputational damage or investor withdrawals. It underscores the point that the path of returns matters as much as the magnitude, and that the best-performing funds are often those with the most disciplined risk management around drawdowns.

Discussion

The findings confirm that drawdowns have a significant and lasting impact on investment manager performance. Severe drawdowns reduce risk-adjusted returns, likely due to behavioral changes such as reduced risk-taking or forced liquidations from redemptions. The positive effect

of risk management policies underscores the importance of stop-loss mechanisms and leverage controls in mitigating drawdown effects.

Manager turnover spikes post-drawdown, reflecting investor pressure and governance mechanisms. However, new managers do not consistently outperform, suggesting that turnover may not always resolve performance issues. Investor redemptions exacerbate performance declines by forcing managers to sell assets at inopportune times, creating a feedback loop of underperformance.

These results have implications for portfolio management. Managers should prioritize robust risk management to limit drawdown severity and maintain investor confidence. Investors, in turn, should consider drawdown recovery patterns when evaluating managers, rather than reacting impulsively to losses.

Limitations include the reliance on historical data, which may not capture future market dynamics, and the potential for unobserved manager characteristics to influence results. Future research could explore the role of investor psychology in redemption decisions or the long-term career impacts of drawdowns on managers.

Conclusion

Drawdowns represent more than a statistical inconvenience; they shape the very fabric of investment manager performance. By constraining capital, altering behavior, and interacting with asymmetric incentives, drawdowns disrupt performance persistence and investor-manager alignment. The evidence suggests that managers who effectively manage drawdowns are more likely to sustain long-term compounding and maintain investor trust.

This study demonstrates that drawdowns significantly impair investment manager performance, increase turnover, and trigger investor redemptions. However, effective risk management can mitigate these effects, enabling faster recovery. Future research should explore the role of investor psychology in redemption decisions and the long-term career impacts of drawdowns on managers. These findings inform both managers and investors navigating the challenges of volatile markets.

The behavioral differences we document between hedge fund and mutual fund managers following drawdowns reflect important structural differences in these investment vehicles. The tendency for hedge fund managers to increase risk following drawdowns, combined with their higher closure rates, suggests that due diligence processes should place particular emphasis on risk management capabilities and organizational stability.

For manager evaluation, our results suggest that severe drawdown events contain predictive information about future manager performance beyond what is captured by traditional risk metrics. Investors may benefit from incorporating drawdown history into their manager selection and monitoring processes, potentially reducing exposure to managers showing early signs of performance deterioration.

For fund managers, our results highlight the importance of robust risk management systems and operational procedures to minimize the probability of severe drawdown events. Given the

persistent nature of post-drawdown underperformance, prevention of severe drawdowns should be prioritized over recovery strategies.

Future research should examine the psychological drivers of investor redemptions and their impact on fund stability. Additionally, longitudinal studies could assess whether drawdowns have lasting effects on manager career trajectories. By understanding the interplay between drawdowns, performance, and behavior, stakeholders can better manage the challenges of investment management.

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